BUS4012 – Vibe Coding for Startups

Assignment 03: Application Development & Peer Review
Assignment Type: Individual Deployment + Peer Review (Group of 4)
Marks: 40%
Due Date: 15th October 2026 11:59 PM (AEST) – submission link on LMS

Building on the MVP you developed in **Assignment 2**, you will now individually develop and deploy a fully functional application using React for the frontend and Python for the backend. Your application must use **a cloud platform Vercel** to host your deployed application, and **Supabase** as your database to persist real user data. Once all group members have deployed their applications, you will act as a peer reviewer and critically evaluating the live applications of each of your three group members from the perspective of an end user, focusing on the application's design, functionality, and overall user experience.

Part A: Individual Development & Deployment

What you are required to do:

Each student is required to independently develop and deploy the application designed in Assignment 2 MVP using the technologies specified below. Your application must include all of the following:

- A **frontend** developed using **React**
- A **backend** developed using **Python**
- A **Supabase** database that stores and retrieves user-submitted data, connected to your backend
- All screens of your application must remain functional, and data entered by users must be saved to and retrieved from Supabase
- If you demonstrate your application running live on a cloud platform in your video, you will receive the **2.5 marks** dedicated for deployment.
- Sensitive credentials such as Supabase API keys and database connection strings must be stored as environment variables and must not be hardcoded or exposed in your frontend code.
- Include all prompts used to develop this application as an appendix to your submission.
- Your project must be committed to **GitHub**. In your demo video, include a walkthrough of your GitHub repository showing how your project is structured and organised.

Part A Deliverable:

A **5-minute demo video** that:

- Showcase your application by opening with the live URL visible in the browser address bar
- Demonstrates the functioning application, including a user completing a full workflow and data being saved to and retrieved from the Supabase database
- Opens the Supabase Table Editor or dashboard to confirm that data submitted through the application is correctly saved to the database

- Briefly explains your application architecture, covering your React frontend, Python backend, and Supabase integration, and demonstrates how sensitive credentials have been managed securely. Include a walkthrough of the key prompts used to develop the application.

- Showcases your **GitHub repository (decide whether to use private or public repository)**, including a brief walkthrough of your repository structure, committed folders, and your **.gitignore** file. This section should be kept concise and no more than 45 seconds.

A prompt document

- Should include main prompts and couple of follow-up prompts used to develop the application, each accompanied by a very brief explanation of what the prompt was intended to achieve.
- Include the GitHub URL of you application in this document.

Part B: Peer Review

What you are required to do:

You will review the deployed applications of each of your **three group members**. For each application, you will produce a structured **written review** that includes a minimum of **5 annotated screenshots** taken from the live deployed application. Your review focuses entirely on the application itself and its design, usability, and functionality from the perspective of an end user. You are not required to access or evaluate the database of your colleague's application.

Screenshots should be captured from the live application. If the application is not hosted, arrange a dedicated time to review the application using your peer's computer.

Note: Given that your peers may or may not deploy their application to a cloud platform, your team is responsible for arranging a time and allowing your application to be reviewed by each group member through your PC or laptop.

Your review should cover the application's purpose and target user, an evaluation of the user interface and visual design (layout, colour, typography, and overall aesthetic), a walkthrough of the application's functionality and user flow, and a security evaluation focusing on the login and authentication flow and how sensitive information is handled in the interface. Provide at least three specific, actionable suggestions for improvement across UI/UX, usability, and any security concerns identified. All feedback must be tied to evidence from your screenshots and written in a respectful, professional tone. **You must include at least five annotated screenshots of the peer's application for each peer review.**

- **Application overview:** Describe the application's purpose, target user, and the problem it aims to solve.
- **User interface and design:** Evaluate the visual design, layout, colour, typography, and overall consistency of the interface.

- **Functionality and user flow:** Walk through the application as an end user and assess whether screens flow logically, buttons and forms work as expected, and navigation is intuitive.
- **Security:** Evaluate the login and authentication flow and how sensitive information is handled in the interface. Consider any other security aspects you observe as an end user.
- **Constructive feedback:** Provide at least three specific, actionable suggestions for improvement covering UI/UX, usability, functionality, and any security concerns identified.

NOTE: You will submit **three separate peer review documents.** (One for each group member's application). In case of if peer didn't provide time for the peer review, do write a reflection your effort in reaching out team lead and relevant team members.

Additionally, the peer details should be clearly specified in the report for each peer review document, including the peer's name, role, communication attempts, and the outcome of the review request.

Deliverables Summary

Deliverable	Marks	Type	Who
5-minute demo video of your developed, document containing the prompts and application	20	Video	Individual
3 × Peer Review documents (min. 5 screenshots each)	20	Written Report	Individual
Total	**40**		

Marking Criteria

Part A: Individual Demo Video (20 marks)

Criteria	Mark s	Pass (P)	Credit (C)	Distinction (B)	High Distinction (A)
Cloud Deployment	2.5	Application is partially or incorrectly deployed. One or more components are missing or not accessible via a live URL.	Frontend and backend are deployed and accessible via a live URL with Supabase connected.	Frontend and backend are fully deployed on a cloud platform with reliable Supabase integration and secure credential management via environment	Seamless and professional deployment of the React frontend and Python backend on a cloud platform. Secure credential management is clearly demonstrated.

				variables, as explained clearly in the demo.	The Supabase integration is robust and the architecture explanation is thorough and confident.
Demo Video Presentation	15	Basic walkthrough showing the live URL and some screens. Limited demonstration of the React/Python/Supabase stack. Explanation lacks structure or clarity.	Clear and structured demonstration of the deployed application covering the React frontend, Python backend, and Supabase integration. Student is visible and communicates clearly.	Engaging and well-paced video confidently demonstrating the full user workflow, React frontend, Python backend, Supabase integration, and cloud deployment.	Highly professional and polished video with confident delivery and clear storytelling. The React and Python architecture, Supabase data flow, and deployment are all explained with strong communication and presentation skills.
Prompt Document	2.5	Prompts are included but explanations are missing or too vague to be meaningful.	Key prompts and follow up prompts are included with brief explanations for most entries.	All prompts and follow up prompts are documented with clear, concise explanations of what each prompt was intended to achieve.	Comprehensive documentation of all prompts and follow-up prompts with insightful explanations that clearly demonstrate the student's vibe coding process and decision-making throughout development.

Part B: Peer Review (20 marks)

Criteria	Marks	Pass (P)	Credit (C)	Distinction (B)	High Distinction (A)
Application Overview & UI Design	6	Overview is vague and screenshots are minimal or missing. UI evaluation lacks specific observations or evidence.	Application purpose and target user are clearly described. At least one UI screenshot is included with basic commentary on design and layout.	Clear and accurate overview with well-chosen screenshots. UI evaluation identifies specific strengths and weaknesses in layout, colour, and visual consistency.	Insightful and thorough overview that sets strong context. UI analysis is detailed and evidence-based, with purposefully annotated screenshots that demonstrate a sophisticated understanding of design principles.
Functionality & User Flow	6	Functionality observations are superficial. Screenshots do not clearly illustrate the user flow or any issues encountered.	Core screens and interactions are evaluated with relevant screenshots. Basic observations about usability and flow are included, though depth may be limited.	Thorough walk-through of the application from an end-user perspective. Annotated screenshots clearly illustrate specific functionality strengths and issues, with logical reasoning provided.	Exceptional evaluation of functionality and user flow. Screenshots are strategically selected and thoroughly annotated. Observations are precise, insightful, and clearly grounded in the end-user experience.
Security Evaluation	4	Security aspects are not evaluated or observations are too vague to be	Basic security observations are made, such as the login flow restricts	Both security aspects are clearly evaluated with supporting screenshots. Observations	Thorough and insightful security evaluation with strong screenshot evidence. The authentication

		meaningful. No screenshots support the security commentary .	access appropriately . Commentary is relevant but lacks depth.	are specific and demonstrate a sound understanding of user-facing security considerations .	flows and other security aspects are assessed with precision, and any concerns identified are clearly articulated and linked to constructive recommendations .
Constructive Feedback	4	Feedback is vague, generic, or not tied to evidence from the application. Fewer than three suggestions are provided.	At least three suggestions for improvement are provided. Feedback is relevant and respectful, though it may lack specificity or direct connection to screenshots.	At least three well-reasoned, specific suggestions that are clearly linked to observations in the review. Feedback is actionable and would genuinely help the developer improve their application.	Highly actionable and evidence-based feedback across UI, usability, functionality, and security. Suggestions are thoughtful, specific, and prioritised in a way that demonstrates genuine critical engagement with the application, written in a professional and constructive tone throughout.